Requirement Specification

REQ-0143 — Recovery — Secure Key Rotation

1. Identification

Requirement ID	REQ-0143
Title	Recovery — Secure Key Rotation
Type	Functional
Priority	High
Severity	Critical
Status	Approved
Version	v1.3
Created	2022-09-23
Updated	2023-06-20
Source Document	SRS-APP-002-v3.pdf

2. Application & Solution

Application ID	APP-002
Application	MobileCommunication
Domain	Telecom
App Version	4.1.1
Solution ID	SOL-015
Solution	Tab Manager
Component	Payment Gateway
Feature	health metric upload

3. Requirement Description

The system shall support firmware over-the-air update without data loss under low-bandwidth conditions.

4. Acceptance Criteria

AC-1: The system shall satisfy the requirement within the defined performance SLA (p95 < 500ms).

AC-2: All failure scenarios must return a human-readable error message to the user.

AC-3: All events must be logged to the audit trail with a valid timestamp and user context.

AC-4: The feature must operate correctly in all supported environments (Mobile, Wearable, Cloud).

AC-5: Security constraints: all data in transit must be encrypted using TLS 1.3 or higher.

5. Dependencies

This requirement depends on the Tab Manager (SOL-015) being operational and at version 4.8 or above.

SYNTHETIC DATA — No real Samsung confidential information

6. Related Test Cases

TC-0185	Test that voice command recognition correctly handles invali...	Performance	Medium	Draft
TC-0418	Confirm that the system recovers gracefully after Resource E...	Performance	Medium	Deprecated
TC-0557	Ensure that user authentication does not exceed the CPU ceil...	Performance	Medium	Active
TC-1583	Verify end-to-end flow of diagnostics report generation from...	Regression	Medium	Draft

7. Related Defects

BUG-0392	UI Failure in Network Stack: ElementNotFoundError: XPat...	Minor	High	FIXED

8. Synthetic Data Notice

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